



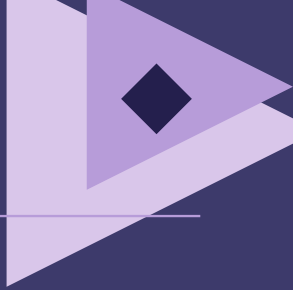
Polar and Euler Form of a Complex Number



Exercise 2.7



Multiplicative Inverse in Polar Form



If $z = r(\cos\theta + i \sin\theta)$, then:

$$z^{-1} = 1/z = 1 / [r(\cos\theta + i \sin\theta)]$$

Multiply by the conjugate: $z^{-1} = (\cos\theta - i \sin\theta) / [r(\cos^2\theta + \sin^2\theta)]$

Since $\sin^2\theta + \cos^2\theta = 1$, this simplifies directly.

Result: $z^{-1} = (1/r) (\cos\theta - i \sin\theta)$



Division of Complex Numbers (Polar Form)

$$z_1 = r_1(\cos\theta_1 + i \sin\theta_1) \quad z_2 = r_2(\cos\theta_2 + i \sin\theta_2)$$

Multiply numerator and denominator by the conjugate of z_2 , then use $i^2 = -1$

Apply: $\cos\theta_1\cos\theta_2 + \sin\theta_1\sin\theta_2 = \cos(\theta_1 - \theta_2)$, $\sin\theta_1\cos\theta_2 - \cos\theta_1\sin\theta_2 = \sin(\theta_1 - \theta_2)$

Result: $z_1/z_2 = (r_1/r_2) [\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)]$

Note: $\arg(z_1/z_2) = \theta_1 - \theta_2 = \arg(z_1) - \arg(z_2)$



Product of Complex Numbers (Property 2)

$$z_1 \cdot z_2 = r_1 r_2 (\cos \theta_1 + i \sin \theta_1)(\cos \theta_2 + i \sin \theta_2)$$

Expand and use $i^2 = -1$ to separate real and imaginary parts

$$\text{Apply: } \cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2 = \cos(\theta_1 + \theta_2), \quad \sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2 = \sin(\theta_1 + \theta_2)$$

$$\text{Result: } z_1 \cdot z_2 = r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$$

$$\text{Note: } \arg(z_1 z_2) = \theta_1 + \theta_2 = \arg(z_1) + \arg(z_2)$$

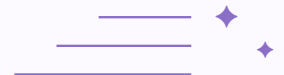




Worked Examples



Applying the Polar Form Properties



Question 1 — Write in Polar Form

Formula: $z = a+ib = r(\cos\theta + i \sin\theta)$, where $r = \sqrt{a^2+b^2}$, $\theta = \tan^{-1}(b/a)$

(i) $2 + i2\sqrt{3}$

$$r = \sqrt{2^2 + (2\sqrt{3})^2} = \sqrt{16} = 4$$
$$\theta = \tan^{-1}(2\sqrt{3}/2) = \pi/3$$

$$z = 4(\cos\pi/3 + i \sin\pi/3)$$

(ii) $3 - i\sqrt{3}$

$$r = \sqrt{3^2 + (-\sqrt{3})^2} = \sqrt{12} = 2\sqrt{3}$$
$$\theta = \tan^{-1}(-\sqrt{3}/3) = -\pi/6$$

$$z = 2\sqrt{3} [\cos(-\pi/6) + i \sin(-\pi/6)]$$

(iii) $-2 - i2$

$$r = \sqrt{(-2)^2 + (-2)^2} = 2\sqrt{2}$$

3rd quadrant $\Rightarrow \theta = \pi + \pi/4 = 5\pi/4$

$$z = 2\sqrt{2} (\cos 5\pi/4 + i \sin 5\pi/4)$$

(iv) $(i-1) / (\cos\pi/3 + i \sin\pi/3)$

$$i-1 = \sqrt{2}(\cos 3\pi/4 + i \sin 3\pi/4)$$

Subtract angles: $3\pi/4 - \pi/3 = 5\pi/12$

$$z = \sqrt{2} (\cos 5\pi/12 + i \sin 5\pi/12)$$

Question 2(i) — Product to Rectangular Form

Given: $(\cos\pi/6 + i \sin\pi/6)(\cos\pi/12 + i \sin\pi/12)$

Step 1

$$(\cos A + i \sin A)(\cos B + i \sin B) = \cos(A+B) + i \sin(A+B)$$

Step 2

$$= \cos(\pi/6 + \pi/12) + i \sin(\pi/6 + \pi/12)$$

Step 3

$$\text{Add angles: } \pi/6 + \pi/12 = 3\pi/12 = \pi/4$$

Final Answer: $\cos(\pi/4) + i \sin(\pi/4)$



Question 2(ii) — Division to Rectangular Form

Given: $[\cos\pi/6 - i \sin\pi/6] / [2(\cos\pi/3 + i \sin\pi/3)]$

Step 1–2

Pull out $\frac{1}{2}$, apply $(1/2)[\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)]$

Step 3

Subtract angles: $\pi/6 - \pi/3 = -\pi/6$

Step 4–5

$\cos(-\pi/6) = \sqrt{3}/2$, $\sin(-\pi/6) = -1/2$; multiply by $\frac{1}{2}$

Final Answer: $\sqrt{3}/4 - i(1/4)$



Question 3 — Modulus Product Identity

Given $(x_1+iy_1)(x_2+iy_2)\cdots(x_n+iy_n) = a+ib$, prove:

$$(x_1^2+y_1^2)(x_2^2+y_2^2)\cdots(x_n^2+y_n^2) = a^2 + b^2$$

1. Take modulus: $|z_1 z_2 \cdots z_n| = |z_1| \cdot |z_2| \cdots |z_n|$

2. Modulus formula: $|x_r+iy_r| = \sqrt{(x_r^2+y_r^2)}$

3. RHS modulus: $|a+ib| = \sqrt{(a^2+b^2)}$

4. Equate and square both sides

Hence Proved



Question 3(ii) — Sum of Arguments Identity

Prove: $\sum_{r=1}^n \tan^{-1}(y_r/x_r) = \tan^{-1}(b/a) + 2k\pi, k \in \mathbb{Z}$

Step 1

$$\arg(z_1 z_2 \cdots z_n) = \arg(z_1) + \arg(z_2) + \cdots + \arg(z_n)$$

Step 2

$$\arg(x_r + iy_r) = \tan^{-1}(y_r/x_r)$$

Step 3

$$\arg(a + ib) = \tan^{-1}(b/a) + 2k\pi$$

Hence Proved: $\sum \tan^{-1}(y_r/x_r) = \tan^{-1}(b/a) + 2k\pi$



Question 4 — Show that $z = i \tan \theta$

Given: $(1+z)/(1-z) = \cos 2\theta + i \sin 2\theta$

$\cos 2\theta + i \sin 2\theta = (1+i \tan \theta)/(1-i \tan \theta)$

Substitute: $(1+z)/(1-z) = (1+i \tan \theta)/(1-i \tan \theta)$

Cross multiply: $(1+z)(1-i \tan \theta) = (1-z)(1+i \tan \theta)$

Expand both sides and cancel the common $-iz \tan \theta$ term

$2z = 2i \tan \theta$

Final Answer: $z = i \tan \theta$



Question 5 — De Moivre's Theorem Application

Given: $\cos\alpha + \cos\beta + \cos\gamma = 0$ and $\sin\alpha + \sin\beta + \sin\gamma = 0$

Part (i)

Show: $\cos 3\alpha + \cos 3\beta + \cos 3\gamma$
 $= 3\cos(\alpha + \beta + \gamma)$

Compare REAL parts after cubing

Part (ii)

Show: $\sin 3\alpha + \sin 3\beta + \sin 3\gamma$
 $= 3\sin(\alpha + \beta + \gamma)$

Compare IMAGINARY parts after cubing

Key idea: $\cos\theta + i\sin\theta = e^{i\theta} \Rightarrow e^{i\alpha} + e^{i\beta} + e^{i\gamma} = 0$

Trick: cube $(e^{i\alpha} + e^{i\beta} + e^{i\gamma})^3 = 0$, then compare real / imaginary parts

$$\cos 3\alpha + \cos 3\beta + \cos 3\gamma = 3\cos(\alpha + \beta + \gamma) \quad | \quad \sin 3\alpha + \sin 3\beta + \sin 3\gamma = 3\sin(\alpha + \beta + \gamma)$$

Question 6 — Argument Condition

Given: $z = x+iy$ and $\arg[(z-i)/(z+2)] = \pi/4$

Show that: $x^2 + y^2 + 3x - 3y + 2 = 0$

Key idea

$$\arg(z_1/z_2) = \arg(z_1) - \arg(z_2)$$

Trick

Argument = $\pi/4 \Rightarrow$ Real part = Imaginary part (after simplification)

Final Answer: $x^2 + y^2 + 3x - 3y + 2 = 0$





Thank You

